

• Path m.t. to calculate $\Psi_{universe}$

$$\langle \mathbb{E}' | e^{-iH\tau} | \mathbb{E} \rangle = \int_{\substack{\varphi(0)=\mathbb{E} \\ \varphi(1-\epsilon)=\mathbb{E}'}} D\varphi e^{iS}$$

Send $(-\tau) \rightarrow (+\infty i + i\epsilon)$

$$\langle \mathbb{E}' | e^{-iH\infty} e^{-H\epsilon} | \mathbb{E} \rangle$$

$$= \int_{\substack{\varphi(0)=\mathbb{E} \\ \varphi(1-\epsilon)=\mathbb{E}'}} D\varphi e^{iS} \propto \Psi_g(\varphi') \Psi_g(\varphi)$$

$$\int D\varphi e^{iS}$$

with $\varphi(-\infty(1-\epsilon)) \rightarrow 0$

then

$$\int D\varphi e^{iS} \propto \Psi_{vac}(\varphi)$$

~~Harmonic oscillator~~: $S = \int dt (\frac{1}{2} \dot{\varphi}^2 - \frac{1}{2} \omega^2 \varphi^2)$

$$Z = \int_{\substack{\varphi(0)=\varphi \\ \varphi(-\infty-i\epsilon) \rightarrow 0}} D\varphi e^{iS + J\varphi}$$

Gaussian: Must also satisfy B.C.'s

classical: $\ddot{\varphi}_{cl} + \omega^2 \varphi_{cl} = 0$
i.e. the E.O.M

or

$$\varphi_{cl} = A e^{i\omega t} + B e^{-i\omega t}$$

Now, for B.C.'s:

$$\begin{aligned} & \varphi_{cl}(-\infty - i\epsilon\infty) \\ & B \exp(-i\omega(-\infty - i\epsilon\infty)) \\ & = B \exp(i\omega\infty) \exp(-\omega\epsilon\infty) \\ & A \exp(i\omega(-\infty - i\epsilon\infty)) \\ & = A \exp(i\omega(-\infty)) \exp(\omega\epsilon\infty) \end{aligned}$$

Blows up

$\therefore B = 0$ $\leftarrow$ Not sure why...

$$\varphi_{cl} = \varphi_0 e^{-i\omega t}$$

to compute $S[\varphi_{cl}]$ note

$$\frac{1}{2} \dot{\varphi}^2 = \frac{1}{2} \omega^2 \varphi^2 = \frac{1}{2} \omega (\dot{\varphi}) \varphi = \frac{1}{2} \dot{\varphi} (\dot{\varphi} + \omega^2 \varphi)$$

$$\int d\Omega \left(\frac{1}{2} \dot{\phi}^2 - \frac{1}{2} \omega^2 \phi^2 \right) = \int d\Omega \left[\frac{1}{2} \mathcal{L}(\dot{\phi}, \phi) - \frac{1}{2} \phi (\dot{\phi}^2 + \omega^2 \phi) \right] \quad \text{E.O.M.}$$

$$S = \frac{1}{2} \int_{-T}^T d\Omega (\dot{\phi}_{ce} \phi_{ce}) = \frac{1}{2} [\dot{\phi}_{ce} \phi_{ce}]_{-T}^T$$

$$\phi_{ce} \rightarrow 0 \quad \text{as} \quad -T \rightarrow -\infty \quad \text{and} \quad T \rightarrow \infty$$

$$S[\phi_{ce}] = \frac{1}{2} \dot{\phi}_{ce}(0) \phi_{ce}(0) = + \frac{i\omega}{2} \phi_0^2$$

$$S[\phi_{ce}] = + \frac{i\omega}{2} \phi_0^2$$

Therefore, the wave function for the harmonic oscillator is given by

$$\psi[\phi] = e^{iS[\phi_{ce}]} = e^{-\frac{1}{2}\omega\phi^2}$$

Free Scalar QFT:

$$\psi[\phi] = e^{-\frac{1}{2}\omega_{\vec{k}} \phi_{\vec{k}} \phi_{-\vec{k}}} \leftarrow \text{summing over } k$$

$$= \exp\left(-\frac{1}{2} \int d^{D-1}k \, \omega_{\vec{k}} \phi_{\vec{k}} \phi_{-\vec{k}}\right)$$

Since $\phi_{-\vec{k}} = \phi_{\vec{k}}^*$ then

$$\psi[\phi] = \exp\left(-\frac{1}{2} \int d^{D-1}k \, \omega_{\vec{k}} |\phi_{\vec{k}}|^2\right)$$

Ground state wave ψ^0

$$\psi[\phi_0, \text{circle with } \vec{k}_0 \text{ and } R] ?$$

$\rightarrow$ We are trying to find the wave ψ^0 of a theory within a sphere of Radius R where ϕ vanishes outside of said sphere.

$$\sim e^{-\phi_0^2 R^{D-2}}$$

How can we realize this? For Scalar QFT we have $\int d^{D-1}k$ in which case we simply

have a term $\sim$
 $\omega_{\vec{k}} = \sqrt{k^2 + m^2}$

So By dim analysis

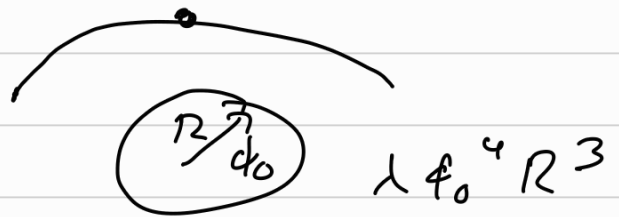
$$\int d^{D-1} k \omega_k \sim \int d^{D-1} k k \quad \left(\frac{D-2}{2} \right): \text{Engineering dim of } \phi$$

$$\delta \phi(R) \sim \frac{1}{R^{(D-2)/2}}$$

Tells us fluctuation of ϕ

$$\left\{ \begin{array}{l} D=1: \sim R^{1/2} \\ \text{Fluctuations increase as } R \rightarrow \infty \\ D=2: \text{ Grow logarithmically} \\ D=3: \text{ "VEV" makes sense at large scales} \end{array} \right.$$

EA: $V(\phi) = \lambda \phi^4$



$$\lambda \phi_0^4 R^3$$

would have to be comparable to

$$\sim \left(\frac{\phi_0^2}{R^2} \right) R^3 \implies \phi_0^2 \sim \frac{1}{R^2 \lambda}$$

χ case
 $\chi(\bigcirc) = e^{-1/\chi}$

Model for General time dep. Harm. oscillator:

$$S = \int dt \left(\frac{1}{2} A(t) \dot{\phi}^2 - \frac{1}{2} B(t) \phi^2 \right)$$

w/ classical E.O.M.

$$\partial_t (A \dot{\phi}_{cl}) + B \phi_{cl} = 0$$

sols: $F_{\pm}(t)$

check $F_{+}(t) \rightarrow e^{i(\text{positive})t}$ as $t \rightarrow -\infty$

$$\phi_{cl} = \phi \frac{F_{+}(t)}{F_{+}(t_0)}$$

$\sim$ F.C.M.

$$= \partial_n (A \dot{\phi}) - \frac{1}{2} \phi (\partial_n (A \dot{\phi}) + B \dot{\phi}) = \mathcal{L}$$

$$e^{iS[\phi_{cl}]} = e^{\frac{i}{2} A_0 \left(\frac{\dot{F}_+(0) \dot{F}_-(0)}{F_+(0)^2} \right) \phi^2}$$

$$\psi(\phi) = \exp \left[\frac{i}{2} A_0 \frac{\partial}{\partial n} (\log(F_+)) \Big|_{n_0} \phi^2 \right]$$

$$|\psi(\phi)|^2 = \exp \left[-\phi^2 \text{Im} \left\{ A_0 \partial_n \log F_+ \Big|_{n_0} \right\} \right]$$

* Propagator/Greens f^μ : Invert $\mathcal{O}(x^2)$ part of action

$$\phi(n) = \int dn' G(n, n') \phi(n')$$

$$\therefore \frac{1}{2} \partial_n (A \dot{\phi}) - \frac{1}{2} B \dot{\phi}^2$$

$$\int \int dn' \phi(n') \left[\partial_n (A G(n, n')) - B G(n', n) \right]$$

$$\therefore \partial_n (A G(n, n')) - B G(n', n) = \delta(n - n')$$

$$\text{Soln} \sim \begin{array}{c} \uparrow n \\ \frac{f_1(n') \quad F_1(n)}{f_2(n') \quad F_2(n)} \end{array} \Rightarrow n' = G(n, n')$$

For $n = n'$: implies $f_1 F_1 = f_2 F_2$

$$\int dn' (\partial_n (A G(n, n')) - B G(n, n')) = 1$$

$$\int dn' A G(n, n') - \int dn' B G(n, n') = 1$$

$$A f_1 \dot{F}_1 - A f_2 \dot{F}_2 = 1 \quad \left(f_2(n) = \frac{f_1 F_1}{F_2}(n) \right)$$

$$(f_1 F_1)$$

$$A(\dot{F}_1 \dot{F}_1 - (\frac{1}{F_2}) \dot{F}_2) = 1$$

$$\dot{F}_1 = \dot{F}_2 / A \{ F_2 \dot{F}_1 - F_1 \dot{F}_2 \}$$

$$\dot{F}_2 = \dot{F}_1 / A \{ F_1 \dot{F}_2 - F_2 \dot{F}_1 \}$$

$$F_1 \dot{F}_2 - F_2 \dot{F}_1 : \text{Wronskian}$$

$$\partial_t (A \dot{F}_1 F_2 - A \dot{F}_2 F_1) = 0$$

So it is a constant
 * Normalise this to be

$$A(\dot{F}_1 F_2 - \dot{F}_2 F_1) \equiv i \quad (\text{choice})$$

$$-i G(n, n') = \Theta(n - n') F_1(n) F_2(n') + \Theta(n' - n) F_2(n') F_1(n) \quad \left(\begin{array}{l} \text{Greens} \\ f_n \end{array} \right)$$

Note, G is not unique. Can shift

$$F_{1,2} \rightarrow F_{1,2} + X(n)$$

* B.C.'s: for $G(n, n')$ not shown in the video

$$(i) n \rightarrow -\infty (1-i\epsilon) \quad G(n, n') \rightarrow 0$$

$$(ii) n \rightarrow 0 \text{ or } n' \rightarrow 0$$

$$G(n, n') \rightarrow 0$$

Computing ψ rather than $\langle, \rangle$ at $n=0$.

$$\Rightarrow -i G(n, n') = \Theta(n - n') F_+(n) F_-(n') + \Theta(n' - n) F_+(n') F_-(n)$$

must add $\left(\frac{F_-}{F_+} \right) F_+(n) F_-(n')$

$$= (F_+ + 10 F_+(n) F_+ + 101)$$

* Harmonic Oscillator: $F_{\pm} = N e^{\pm i \omega n}$

Wronskian: $\dot{F}_+ F_- - \dot{F}_- F_+ = 2i N^2 \omega = i$
 $N = 1/\sqrt{2\omega}$

$$G(n, n') = \Theta(n - n') e^{i \omega n} e^{-i \omega n'} \\ + \Theta(n' - n) e^{i \omega n'} e^{-i \omega n} \\ = e^{i \omega n} e^{i \omega n'}$$

ΔS_{cl} : $S = \int d^4 x \sqrt{g} \left(-\frac{1}{2} g^{\mu\nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 \phi^2 \right)$

$$g_{\mu\nu} dx^{\mu} dx^{\nu} = ds^2 = dt^2 - e^{2Ht} d\vec{x}^2$$

$$= \frac{1}{\eta^2} (d\eta^2 - d\vec{x}^2)$$

$$\eta = e^{-t}, \quad H = 1$$